

Assignment_Module_01.py

```
1  # =====
2  # PROJECT : STUDENT RESULT MANAGEMENT SYSTEM
3  # =====
4
5  # -----
6  # 1. Student Information
7  # -----
8  student_name = input("Enter Student Name: ")
9  student_id = input("Enter Student ID: ")
10 department = input("Enter Department: ")
11
12 print()
13
14 # -----
15 # 2. Subject Marks
16 # -----
17 # declaring the subjects list:
18 subjects = ["Python", "Math", "English", "Physics", "ICT"]
19
20 marks = {}          # initiating an empty dictionary named marks
21
22 print("Enter Marks:")
23
24 for subject in subjects:
25     mark = float(input(f"Marks obtained in {subject}: "))
26     marks[subject] = mark      # Dictionary Syntax : dictionary[key] = value
27     # here, subject = key, mark = value
28 print()
29
30 # -----
31 # 3. Calculate Result
32 # -----
33 total = sum(marks.values())
34 average = total / len(marks)
35 highest = max(marks.values())
36 lowest = min(marks.values())
37
38 # -----
39 # 4. Grade Calculation
40 # -----
41 if average >= 80 and average <= 100:
42     grade = "A+"
43 elif average >= 70 and average <= 79:
44     grade = "A"
45 elif average >= 60 and average <= 69:
46     grade = "A-"
47 elif average >= 50 and average <= 59:
48     grade = "B"
49 elif average >= 40 and average <= 49:
50     grade = "C"
51 else:
52     grade = "F"
53 # Easy alternative :
54 # -----
55 # if average >= 80:
56 #     grade = "A+"
57 # elif average >= 70:
58 #     grade = "A"
59 # elif average >= 60:
60 #     grade = "A-"
61 # elif average >= 50:
62 #     grade = "B"
63 # elif average >= 40:
64 #     grade = "C"
65 # else:
```

```

66 #     grade = "F"
67
68 # -----
69 # 5. Pass or Fail
70 # -----
71 status = "Passed"
72
73 for mark in marks.values():
74     if mark < 40:
75         status = "Failed"
76         break
77 # It means:
78 #     Assume the student has Passed.
79 #     Check each subject mark.
80 #     If any mark is below 40, change the status to Failed and stop checking.
81
82 # -----
83 # 6. Password Verification
84 # -----
85 correct_password = "python123"
86
87 password = input("\nEnter Password: ")
88
89 while password != correct_password:
90     print("Incorrect Password! Try Again.")
91     password = input("Enter Password: ")
92
93 print("Password Verified Successfully!")
94
95 # -----
96 # 7. String Operations
97 # -----
98 print("\n===== STRING OPERATIONS =====")
99 print("Uppercase      :", student_name.upper())
100 print("Lowercase      :", student_name.lower())
101 print("Length          :", len(student_name))
102 print("First 3 Letters :", student_name[:3])
103 print("Last 3 Letters  :", student_name[-3:])
104
105 # -----
106 # 8. Set Example
107 # -----
108 sports = {"Football", "Cricket", "Badminton"}
109 clubs = {"Programming", "Cricket", "Photography"}
110
111 print("\n===== SET OPERATIONS =====")
112 print("Common Items :", sports.intersection(clubs))
113 print("All Unique Items :", sports.union(clubs))
114
115 # -----
116 # 9. Tuple Example
117 # -----
118 weekdays = (
119     "Sunday",
120     "Monday",
121     "Tuesday",
122     "Wednesday",
123     "Thursday",
124     "Friday",
125     "Saturday"
126 )
127
128 print("\n===== TUPLE EXAMPLE =====")
129 print("First Day :", weekdays[0])
130 print("Last Day  :", weekdays[-1])
131 print("Total Days:", len(weekdays))
132

```

```

133 # -----
134 # 10. Final Report
135 # -----
136 print("\n" + "=" * 35)
137 print("          STUDENT REPORT")
138 print("=" * 35)
139
140 print(f"Name       : {student_name}")
141 print(f"ID        : {student_id}")
142 print(f"Department : {department}")
143
144 print("-" * 35)
145
146 for subject, mark in marks.items():
147     print(f"{subject:<10} : {mark}")
148 # {subject:<10} meaning :
149 # 'subject' prints the value of the variable
150 # '<' means left-align the text
151 # '10' means use a field width of 10 characters.
152
153 # Alternative code :
154 #-----
155 # for subject in marks:
156 #     print(subject, ":", marks[subject])
157
158 # Explanation :
159 # -----
160 # To access a value from a dictionary, we write
161 # dictionary[key]
162 # here, marks[subject]
163 # Python looks for the key--> subject
164 # in the dictionary--> marks and
165 # returns its value.
166
167
168 print("-" * 35)
169
170 print(f"Total       : {total}")
171 print(f"Average    : {average:.2f}")
172 # ':' means "I want to format this value"
173 # '.2' means "Show 2 digits after the decimal point"
174 # 'f' means "Display the number as a floating-point number."
175 print(f"Highest    : {highest}")
176 print(f"Lowest     : {lowest}")
177 print(f"Grade      : {grade}")
178 print(f"Status     : {status}")
179
180 print("=" * 35)

```